Torsion Collapse for Noncommutative Cotlar Symbols

Automatic math-discovery packet

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Source question

The source paper is Adrian M. Gonzalez-Perez, Javier Parcet, and Runlian Xia, *Noncommutative Cotlar identities for groups acting on tree-like structures*, arXiv:2209.05298. The paper proves that, for a discrete group and $G_0 = \{e\}$, the noncommutative Cotlar identity for a Fourier multiplier is equivalent to the closed formula

$$\big(m(g^{-1}) - m(h)\big)\big(m(gh) - m(g)\big) = 0, \qquad g \neq e,\ h \in G. \tag{1}$$

After proving that UAC-space models produce such symbols, the source asks whether there are Cotlar symbols that do not arise from the UAC models, or whether a reverse construction is always possible.

characteristic less than -1 acts freely on an $\mathbb{R}$-tree [44].

On the other hand, there are many examples of groups for which any action on an $\mathbb{R}$-tree has a global fixed point. When this happens the group G is said to have property (F$\mathbb{R}$), see [3, 59]. The following corollary of Proposition 2.4 characterizes groups with property (F$\mathbb{R}$).

Corollary 2.5. G *has property* (F$\mathbb{R}$) *if and only if the symbol* m *in Model 2 is trivial for any* $x_0 \in X$ *in any* $\mathbb{R}$*-tree* X *on which* G *acts (isometrically).*

One natural question, that we leave open, is whether there are functions $m : \mathrm{G} \to \mathbb{C}$ satisfying (Cotlar) for $\mathrm{G}_0 = \{e\}$ and such that they do not lift to a function $\widetilde{m}$ on a UAC space like in Model 1 or 2 on which G acts. One way to prove the non-existence of such lift $\widetilde{m}$ would be a procedure to assemble from the group G and the function m a G-space X on which G acts naturally and a lift $\widetilde{m} : X \to \mathbb{C}$ satisfying hypothesis like those of Models 1 and 2. So far, this reverse construction has escaped us.

3. Left orderable groups

Recall that a *total* or *linear* order $\preceq$ is an order relation such that, given any two points x, y then it holds that $x \preceq y$ or $y \preceq x$. A *left-orderable group* is a group admitting a left-invariant total order, ie $(\mathrm{G}, \preceq)$ satisfies that $g \preceq h$ if and only if $kg \preceq kh$. Recall that, as we defined in the introduction, every left-orderable group has a *sign function* $\mathrm{sgn} : \mathrm{G} \to \mathbb{C}$ that assigns $+1$ or -1 depending on whether $e \prec g$ or $g \prec e$. We will prove that $H = T$ is bounded

Figure 1: The source's reverse-UAC question, on page 21 of the arXiv PDF.

Packet claim

This packet proves a torsion obstruction. If G is a torsion group, then (1) forces m to be constant away from the identity. Thus a nontrivial counterexample to the reverse-UAC question cannot live on a torsion group. The remaining constant-off-identity symbols are geometrically uninteresting: they lift to Model 1 on a regular star tree.

Proof intuition

The Cotlar formula is very rigid on finite cycles. If a has finite order and $m(a) \neq m(a^{-1})$, then applying (1) repeatedly with $g = a$ walks around the cyclic subgroup and forces every negative power of a to have the value $m(a^{-1})$, eventually including a itself, a contradiction. Thus finite-order elements have the same value as their inverses.

Now suppose two torsion elements $a, b \neq e$ have different values. Applying (1) to $g = a$, $h = a^{-1}b$ forces $a^{-1}b$ to have the same value as a. Applying it to $g = b$, $h = b^{-1}a$ forces $(a^{-1}b)^{-1}$ to have the same value as b. Since $a^{-1}b$ also has finite order, inverse-symmetry gives a contradiction.

Main theorem

Lemma 1 (finite-order inverse symmetry)**.** *Let G be a group and let $m : G \to \mathbb{C}$ satisfy* (1)*. If $a \in G \setminus \{e\}$ has finite order, then*

$$m(a) = m(a^{-1}).$$

Proof. If $a^2 = e$, the assertion is immediate. Assume that the order of a is $q \geq 3$, and suppose for contradiction that $m(a) \neq m(a^{-1})$. We prove by induction that

$$m(a^{-j}) = m(a^{-1}) \qquad (1 \leq j \leq q-1).$$

The case $j = 1$ is tautological. Suppose the claim holds for some $1 \leq j \leq q-2$. Apply (1) with $g = a$ and $h = a^{-(j+1)}$. Since $gh = a^{-j}$, we get

$$\bigl(m(a^{-1}) - m(a^{-(j+1)})\bigr)\bigl(m(a^{-j}) - m(a)\bigr) = 0.$$

By the induction hypothesis and by $m(a^{-1}) \neq m(a)$, the second factor is nonzero. Hence $m(a^{-(j+1)}) = m(a^{-1})$. Taking $j = q-1$ gives

$$m(a) = m(a^{-(q-1)}) = m(a^{-1}),$$

contradicting the assumption. Therefore $m(a) = m(a^{-1})$. □

Theorem 1 (torsion Cotlar collapse)**.** *Let G be a torsion group and let $m : G \to \mathbb{C}$ satisfy* (1)*. Then there is a scalar $\alpha \in \mathbb{C}$ such that*

$$m(g) = \alpha \qquad \textit{for every } g \in G \setminus \{e\}.$$

Proof. Assume that there are $a, b \in G \setminus \{e\}$ with $m(a) \neq m(b)$. Then $a \neq b$, so $c = a^{-1}b \neq e$. Since G is torsion, a, b, and c all have finite order.

Apply (1) with $g = a$ and $h = c = a^{-1}b$. Since $gh = b$, we obtain

$$\bigl(m(a^{-1}) - m(c)\bigr)\bigl(m(b) - m(a)\bigr) = 0.$$

The second factor is nonzero, so Lemma 1 gives

$$m(c) = m(a^{-1}) = m(a).$$

Similarly, applying (1) with $g = b$ and $h = b^{-1}a = c^{-1}$ gives

$$m(c^{-1}) = m(b^{-1}) = m(b).$$

But c has finite order, so Lemma 1 also gives

$$m(c) = m(c^{-1}).$$

Thus $m(a) = m(b)$, a contradiction. Hence all nonidentity elements have the same value. □

Consequence for the reverse-UAC question

Corollary 1. *No torsion group can provide a nonconstant Cotlar symbol for the reverse-UAC question with $G_0 = \{e\}$. More precisely, every Cotlar symbol on a torsion group is constant off the identity and has a Model 1 lift on a UAC space.*

Proof. The first assertion is Theorem 1. It remains to describe the lift for a symbol with

$$m(e) = \beta, \qquad m(g) = \alpha \quad (g \neq e).$$

Let X be the star tree with one branch $[0,1]_r$ for each $r \in G$, all glued at 0. Let G act by permuting branches:

$$k \cdot (t, r) = (t, kr).$$

Choose $x_0 = (1, e)$. Then $\mathrm{Stab}(x_0) = \{e\}$, and $X \setminus \{x_0\}$ is arcwise connected. Define

$$\widetilde{m}(x_0) = \beta, \qquad \widetilde{m}(x) = \alpha \quad (x \neq x_0).$$

This function is constant along arcs in $X \setminus \{x_0\}$ and is invariant under the stabilizer of x_0. Therefore it is a Model 1 lift, and

$$\widetilde{m}(g \cdot x_0) = m(g)$$

for every $g \in G$. □

Scope and limitations

This is not a full solution of the reverse-UAC problem. It shows that a nontrivial counterexample must use group elements of infinite order. More locally, the proof shows that if $a, b, a^{-1}b$ all have finite order, then any Cotlar symbol satisfying (1) must have $m(a) = m(b)$.

The theorem also does not address the convex-hull problem for Cotlar-type multipliers or the endpoint-space problem from the same source paper.

Novelty and search status

This packet was produced as a direct attack on the source question. Bounded searches were run for phrases including “Cotlar torsion group constant Fourier multiplier”, “Noncommutative Cotlar identities torsion”, and “2209.05298 torsion Cotlar”, and for the source title together with “reverse construction” and “UAC”. The searches found the source arXiv page but no prior statement of the torsion-collapse lemma. This is a partial novelty check only; the result should be reviewed as a short algebraic observation adjacent to the source question.

Computational sanity check

The script `code/finite_cotlar_search.py` exhausts all three-color equality patterns on a few small finite groups, including cyclic groups, dihedral groups, S_3, and elementary 2-groups of ranks 2 and 3. It finds no nonconstant-off-identity patterns. The proof above does not rely on this computation.

Verification notes

The proof uses only the closed Cotlar formula (1). The induction in Lemma 1 is the only cyclic argument: once $m(a^{-j}) = m(a^{-1}) \neq m(a)$, the Cotlar formula with $g = a$, $h = a^{-(j+1)}$ forces the next negative power to have the same value. Theorem 1 then uses the finite order of $a^{-1}b$; this is exactly where the torsion hypothesis enters.

Human-review recommendation

Review as a likely valid partial result. The key checks are Lemma 1, the use of $c = a^{-1}b$ in Theorem 1, and the claim that the regular-star tree gives a legitimate Model 1 lift for the constant-off-identity case.

References

[1] A. M. Gonzalez-Perez, J. Parcet, and R. Xia, *Noncommutative Cotlar identities for groups acting on tree-like structures*, arXiv:2209.05298.